👩‍🏫 **Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

**UNIT 2 · STANDARD 6.DS.1**

# Understand Statistical Questions

Lesson 2-1 · Teacher Copy

**Name:** _____ **Date:** _____ **Class:** _____

**FRONT OFFICE MISSION**

## Scout the Hardwood

You are the new data analyst for the Riverside Hoops youth league. The head coach drops a stack of questions on your desk — some will yield rich data full of variety, others just one fixed fact. Master statistical questions and you decide what the front office actually investigates.

## My Notes

**Level 3 Enrichment**

🎯 **TODAY'S OBJECTIVES**

**Content Objective: I can tell the difference between a statistical question and a non-statistical question.**

**Language Objective: I can explain my reasoning using the words statistical question, data, variability, and survey.**

📖 See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.

✏️ **Fill in each blank as we go. Use the Word Bank to help you.**

**WORD BANK — FILL EACH BLANK WITH THE BEST WORD**

**Statistical Question** | **Non-statistical Question** | **Data** | **Variability** | **Survey** | **Data distribution** | **Dot plot**

*Level 3: try the blanks from memory first, then check the bank.*

1. A question that expects a variety of answers and is answered by collecting data is a _____ .

2. A question that does not anticipate variability in the data is a _____ .

3. Facts or numbers collected to answer a question are _____ .

4. How much the values in a data set differ from one another is the _____ .

5. A way to collect data by asking many people the same question is a _____ .

6. The way data values are spread out across a range is the _____ .

7. A number line that shows one dot for each data value is a _____ .

## Write About the Math

**Which question is statistical? Why would the editor's question give the newspaper more interesting data?**
*¿Qué pregunta es estadística? ¿Por qué la pregunta de la editora le daría al periódico datos más interesantes?*

Say your answer to a partner first. Then write it, using at least two of these words:

- ☐ **Statistical Question**
  *Pregunta estadística*
- ☐ **Non-statistical Question**
  *Pregunta no estadística*
- ☐ **Data**
  *Datos*
- ☐ **Variability**
  *Variabilidad*
- ☐ **Survey**
  *Encuesta*

**Model:** *A statistical question expects answers that vary across many cases.* — Students identify "How tall is each player?" as statistical because heights vary, while "How tall is our point guard?" has one fixed answer.

### Start

Complete the frames. Use the word bank.
*Completa las oraciones. Usa el banco de palabras.*

- **The reporter's question is ___ because ___. The editor's question is ___ because ___. The editor's question is better for a story because ___.**
- **My answer is ___ because ___.**
  *Mi respuesta es ___ porque ___.*

### Build

Write two connected sentences. Use because, so, or but.

*Escribe dos oraciones conectadas. Usa porque, entonces o pero.*

- **My claim is ___.**
  *Mi afirmación es ___.*
- **My evidence is ___, so ___.**
  *Mi evidencia es ___, entonces ___.*

### Explain

Write a claim, give math evidence, and explain your reasoning.

*Escribe una afirmación, da evidencia matemática y explica tu razonamiento.*

- [ ] I answered the question.
- [ ] I used math evidence: numbers, an equation, or a model.
- [ ] I used at least two math words.

## Writing Guide — Teacher Copy

- The answer to the math question is correct.
- The evidence is real lesson mathematics (numbers, equation, or model), not restated claim.

- The support level changed the language scaffolding, never the mathematical expectation.

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## Answer Key & Teacher Guide

1. **Notes 1:** Statistical Question
2. **Notes 2:** Non-statistical Question
3. **Notes 3:** Data
4. **Notes 4:** Variability
5. **Notes 5:** Survey
6. **Notes 6:** Data distribution
7. **Notes 7:** Dot plot
8. **Watch for this mistake:** *A common mistake in Statistical Questions and Data is judging a question by its topic or its numbers instead of its variability. Students see a question about ONE player, ONE game, or ONE fixed fact — like "How many points does the team need to win a game?" — and call it statistical just because it has a number or is about sports. Or they see a question about a single player's whole season, like "How many touchdowns did the quarterback throw this season?", and call it NOT statistical because only one player is named, forgetting that a season covers many games with a different result each time. Both errors skip the real test: will the answers actually differ from case to case (game to game, player to player)? If yes, it is a statistical question; if there is one fixed answer, it is not.*
9. **Try It 1:** C. How many rebounds does each player grab per game? — *"How many rebounds does each player grab per game?" anticipates variability — different players grab different numbers of rebounds, so you must collect data. The other three each have one fixed answer.*
10. **Try It 2:** B. Because the warm-up times will vary from one athlete to another — *A statistical question anticipates variability. Different athletes warm up for different amounts of time, so the data you collect will vary.*